Is the world supply of omega-3 fatty acids adequate for optimal human nutrition?

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Purpose of review
To delineate the available sources of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) for human consumption and to determine if the available supply is capable of supplying the nutrient levels recommended by expert bodies.

Recent findings
There are converging opinions among experts, professional organizations and health professionals that a recommendation for a daily individual consumption of 500 mg of EPA/DHA would provide health benefits, and this translates to an annual human consumption of 1.3 million metric tons. Current human consumption of EPA/DHA is estimated to be only a small fraction of this amount and many people may suffer from suboptimal health as a result of low intake. EPA and DHA originate in the phytoplankton and are made available in the human food chain mainly through fish and other seafood.

Summary
The fish catch is not elastic and in fact has long since reached a plateau. Aquaculture has grown rapidly, but most of the fish oil produced is currently being used to support aquaculture feed and so this would appear to limit aquaculture growth – or at least the growth in availability of fish sources of EPA/DHA. Vegetable oil-derived alpha-linolenic acid, though relatively plentiful, is converted only at a trace level in humans to DHA and not very efficiently to EPA, and so cannot fill this gap. Microbial EPA/DHA production can in the future be increased, although this oil is likely to remain more expensive than fish oil. Plant sources of EPA and DHA have now been produced in the laboratory via transgenic means and will eventually clear regulatory hurdles for commercialization, but societal acceptance remains in question. The purpose of this review is to discuss the various sources of omega-3 fatty acids within the context of the potential world demand for these nutrients. In summary, it is concluded that fish and vegetable oil sources will not be adequate to meet future needs, but that algal oil and terrestrial plants modified genetically to produce EPA and DHA could provide for the increased world demand.

Keywords
fish oil availability, microbial eicosapentaenoic acid/docosahexaenoic acid production, supply of omega-3 fatty acids

INTRODUCTION

There is a robust literature suggesting the health benefits of the long-chain omega-3 fatty acids, eicosapentaenoic acid (EPA, 20 : 5n3) and docosahexaenoic acid (DHA, 22 : 6n3), for human use. Many different recommendations have been made for the daily intakes of these nutrients, as will be discussed below, but all have agreed that median levels of intake in the Western world should be increased. Nevertheless, it must also be stated that official bodies such as the (US) Institute of Medicine or the USDA have not promulgated a daily recommended intake (DRI) or even adequate intake values for EPA or DHA. Recently, the US Food and Drug Administration (FDA) announced that EPA and DHA have been selected as top priority nutrients for DRI reviews based on public health and policy importance. It may be anticipated that such recommendations will be made in the future and that this will lead to an increased demand for these nutrients. The context for the present analysis will therefore be

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KEY POINTS

- Many expert bodies have recommended increased intakes of the omega-3 essential fatty acids, EPA and DHA, at a level of 500 mg/day for optimal nutrition.
- For about 7 billion humans, this would equate to a current demand of about 1.3 million metric tons of EPA/DHA.
- The present fish/aquatic sources of EPA/DHA can supply about 0.2 million metric tons for human consumption.
- The vegetable oil omega-3 ALA cannot support increased levels of human DHA.
- The large gap in omega-3 nutrition security may be made up by greatly increased algal oil fermentation or in the near future by vegetable oil sources after suitable genetic modification.

the recommended levels of dietary intake for these essential fatty acids as prescribed by various expert bodies. An estimated worldwide demand for EPA and DHA can then be calculated for the world population. Various sources of these fatty acids will then be considered in order to estimate the current supply and to evaluate when necessary growth that can be realized in the near future, also taking development of new technologies into account. It will be necessary to estimate both fish catch and fish oil annual production, as well as to consider algal, krill and other additions to the EPA/DHA supply. Additionally, alpha-linolenic acid (ALA, 18 : 3n3) production from terrestrial plant sources will be considered since it is the metabolic precursor to EPA and DHA in conjunction with the ability of human beings to perform this metabolism. Finally, possible solutions to projected inadequacies in the EPA/DHA supply will be considered, including expanded algal production and genetically modified plant sources.

EPA/DHA INTAKE RECOMMENDATIONS FROM EXPERT BODIES

Many expert bodies in various countries have made statements concerning the recommended intake [1–9] of the nutrients EPA/DHA, generally as a total of these two fatty acids since they generally occur together in food sources. These recommendations have been discussed in more detail recently [10]. Examination of a list of such recommendations (Table 1) indicates that most recommendations are in the range of 250–1000 mg/day. In the USA, two of the most prominent dietary recommendations are from the American Dietetic Association [11] and the American Heart Association, which recommend an intake of about 500 mg/day. Both the World Health Organization/Food and Agriculture Organization (WHO/FAO) and European Food Safety Authority (EFSA) recommend 250 mg EPA/DHA daily intake, whereas the Japanese Ministry for Health opts for 1000 mg/day, either as a capsule or daily fish intake. One may consider a median expert body recommendation to be 500 mg EPA/DHA per day. In contrast to these recommendations, however, the daily intake of EPA/DHA in most of the world is considerably lower than 500 mg/day. It seems that only Japan, some Scandinavian countries and some coastal areas where fish is plentiful have median intakes that are greater than this level. Thus the individual intake would be 365 days × 0.5 g/day or 182.5 g/year. For a world

Table 1. Omega-3 PUFA intake recommendations of expert international bodies for adults

Expert body	Year	Target population	Daily recommendation	Reference
American Heart Association	2011	Heart health	Two fish meals for primary protection	[1]
Heart Foundation Australia	2008	Heart health	500 mg EPA/DHA for primary prevention	[2]
FAO/WHO Expert Consultation	2010	General health	250 mg EPA/DHA	[3]
European Food Safety Authority	2010	General health	250 mg EPA/DHA	[4]
Japanese Ministry of Health	2009	General health	>1 g EPA/DHA	[5]
Health Council Netherlands	2006	General health	450 mg EPA/DHA from fish	[6]
Australia New Zealand National Health and Medical Research Council	2006	Chronic disease	n-3 LC-PUFAs: 610 mg for men 430 mg for women	[7]
Belgian Superior Health Council	2009	Heart health	Daily fatty fish or 1 g capsule	[8]
Agence Francais de Securite Sanitaire des Aliments	2014	General health	500 mg EPA/DHA	[9]

DHA, docosahexaenoic acid; EPA, eicosapentaenoic acid; LC, long chain; PUFA, polyunsaturated fatty acid.

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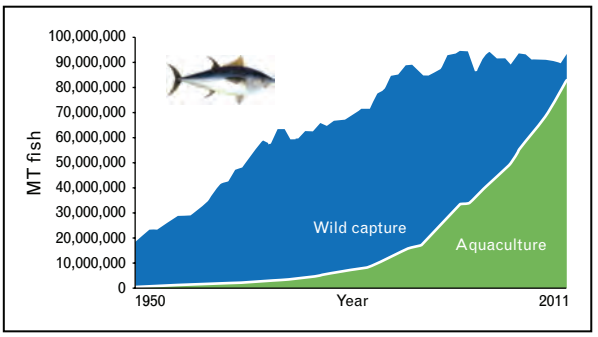


FIGURE 1. Global fish production. Rapid rise of aquaculture and plateauing of wild fish catch. Data from [12].

population of 7.2 billion people, the total global intake requirement would be 182.5 g/year × 7.2 billion or 1314 billion g/year. This equates to about 1.3 billion kg/year or 1.3 million metric tons of EPA/DHA per annum.

FISH/MICROORGANISM SOURCES OF EPA/DHA

Now that we have an estimate of how much EPA/DHA is required, let us turn to the issue of where we can obtain these essential fats and what quantities are available from different sources. EPA, DHA and other long-chain n-3 fatty acids are biosynthesized mainly by phytoplankton, which are consumed by zooplankton and then enter the fish food chain. Sources of dietary EPA/DHA for humans are principally composed of fish and fish oil. Figure 1 depicts the yearly fish catch since 1950 as well as the rapid increase in fish obtained through aquaculture. These data are taken from FAO FishStat and a report of these data is also available [12]. The wild capture of fish peaked in the early 1990s and has been fairly constant since at about 90 000 000 metric tons of fish. The yield of fish via aquaculture has grown linearly since the early 1990s and has now reached nearly the same level as the wild fish catch. As sources for EPA/DHA, anchovies are the largest single species of wild fish and much of this is obtained from the Peruvian fishery. Menhaden, cod, whiting, carp, mackerel, tuna, salmon, pollock, capelin and sardine, in roughly that order, are significant species of fish harvested (Fig. 2). Much of the anchovy catch is consumed in the extraction fishery for fish oil and fish protein, and so on. Our best estimates of the total EPA/DHA produced is about 530 000 metric tons and this estimate comes from the fish industry trade association – the Global Organization for EPA and DHA omega-3s (GOED) (Fig. 3). The portions of this that contribute to human usage break down as follows:

(1) 125 000 metric tons consumed directly from seafood

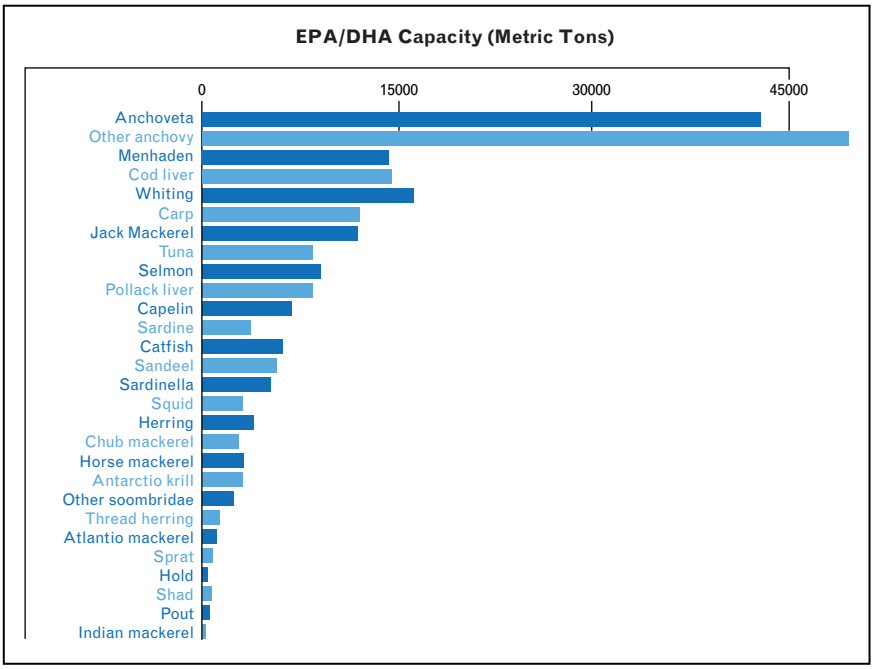


FIGURE 2. Major fish species which contribute to EPA/DHA production. Data compiled by the Global Organization for EPA and DHA Omega-3s (GOED) and based on a calculation of FAO FishStat data [12]. DHA, docosahexaenoic acid; EPA, eicosapentaenoic acid.

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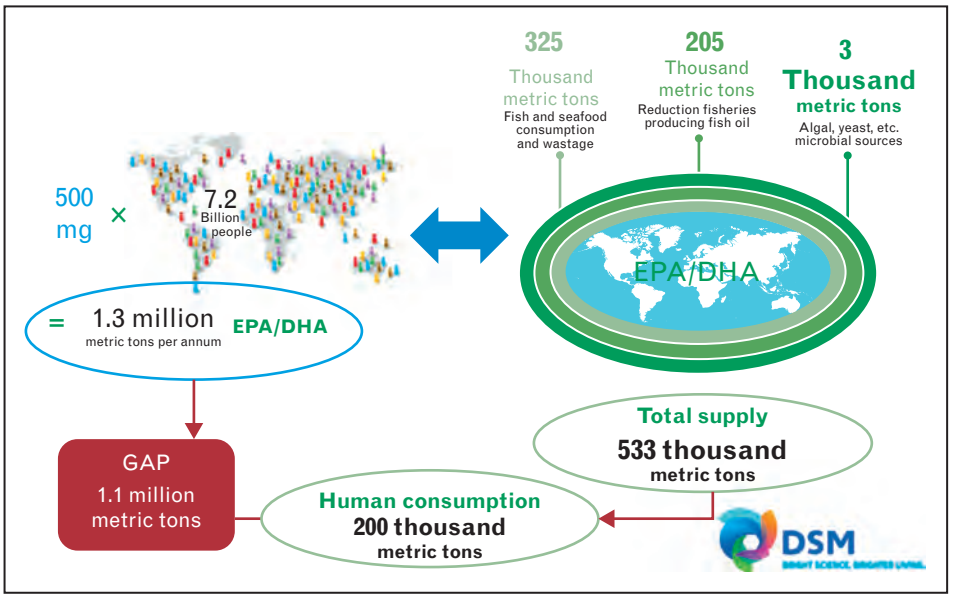


FIGURE 3. Summary slide of worldwide EPA/DHA requirement and available supplies for human consumption. Source is DSM Nutritional Products, *Nutrition Science & Advocacy, 2014*. DHA, docosahexaenoic acid; EPA, eicosapentaenoic acid.

(2) 9000 metric tons utilized by humans of the 200 000 tons derived from seafood industry cuttings
(3) 63 000 metric tons utilized by humans of the 205 000 metric tons from fish oil extraction
(4) 197 000 metric tons total human use from fish sources
(5) 3000 metric tons from nonfish sources including algae, yeast, krill, and so on
(6) 200 000 metric tons EPA/DHA grand total use

The total is already likely an overestimate since it assumes that all of the seafood is consumed, although food wastage is frequently one-third or more. Very little (less than 5%) of the seafood cuttings total is utilized for human consumption and would appear to present an opportunity to significantly increase its use. However, such processing is done in a decentralized fashion such that there is little economic incentive to process the oil appropriately for this purpose. In extraction fisheries, 205 000 metric tons of fish oil are produced, and most of it goes to supply the aquaculture industry with a source of EPA/DHA since salmonids and other fish require this nutrient for optimal growth [13]. Nonfish sources of EPA/DHA come mainly from fermented algal oils and secondarily from Krill oil, but at present contribute a very small percentage of the total available EPA/DHA. The total EPA/DHA available for human consumption (200 000 metric tons) represents only about 15% of the above calculated human requirement (1.3 million metric tons).

ALPHA-LINOLENIC ACID AS A SOURCE OF EPA/DHA

The vegetable oil source of omega-3 fatty acids as ALA has been proposed as another possible source of EPA and DHA. Vegetable oils are relatively abundant and Table 2 lists the United States Department of Agriculture (USDA) estimates of major global production in 2013–2014 [14]. Soybean and rapeseed oil are the major sources of ALA, and the total calculated ALA produced this year is estimated to be over 6100 metric tons. Flax oil (not included in the USDA estimate) is high in ALA content, with a range from about 40 to 60% of fatty acids, and worldwide production of flax seed was estimated in 2010 to be about 525 thousand metric tons per year [15]. The seed is about 23 wt% ALA and this then yields an additional 121 thousand metric tons of worldwide ALA production. The total worldwide ALA production from vegetable oils can therefore be estimated to be about 6220 metric tons.

Contributions of ALA, however, as a source of EPA/DHA presupposes that humans can efficiently metabolize the ALA to EPA and DHA. The rate of this conversion may be examined in two ways: by feeding humans ALA and measuring any increases in bloodstream levels of EPA and DHA, or by determining endogenous human ALA metabolism *in vivo* through metabolic tracers. Pawlosky and Salem performed a tracer experiment wherein deuterated-ALA was given to human adults on a controlled Western diet and in vivo metabolism monitored via the bloodstream, reviewed in [16]. These authors, working at the National Institutes of Health, concluded

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Table 2. USDA estimate of global major vegetable oil and calculation of alpha-linolenic acid production in 2013–2014

Global production of oils	Oil quantity (millions of metric tons)	Percentage of ALA	Quantity of ALA (thousands of metric tons)
Coconut	3.74	0.1	3.7
Cottonseed	5.25	0.1	5.2
Olive	3.28	0.6	19.7
Palm	58.08	0.3	174.2
Palm Kernel	6.8	0.2	13.6
Peanut	5.32	0.3	16.0
Rapeseed	24.48	9.6	2350.1
Soybean	44.62	7.8	3480.4
Sunflower	14.89	0.3	44.7
Totals	166.44	–	6108

ALA, alpha-linolenic acid.
From: http://www.ers.usda.gov/data-products/oil-crops-yearbook.aspx [14].

that only about 0.2% of the absorbed plasma ALA was destined for EPA synthesis. Moreover, less than 0.05% of the plasma ALA was destined for DHA synthesis. This is consistent with many human supplementation studies in which large doses of ALA are given, and the bloodstream shows no change in DHA and generally significant but small increases in EPA [17]. Representatives of the International Society for the Study of Fatty Acids and Lipids – a group of international experts of lipid scientists – have published a position paper concerning ALA metabolism, indicating that neither dietary ALA (see [17] for review, Table 1) nor EPA (see [17] for review, Table 2) is a good source of DHA in humans due to the extremely low level of in-vivo metabolism [16,17]. ALA can support a higher level of bloodstream EPA, but is not nearly as effective as is preformed EPA in this respect. EPA supplementation is also not able to increase the level of DHA in the human bloodstream [17,18].

A recent study (Table 3) in which ALA, EPA or DHA was supplemented for 6 weeks to humans confirms these conclusions [18]. The ALA supplementation produced a very small (0.24%) increase in erythrocyte ALA, a nonsignificant increase (0.12%) in EPA and a slight decrease in DHA. The EPA-supplemented group exhibited only a significant increase in EPA, which was much more substantial (4.41%). DHA supplementation produced significantly increased EPA (1.56%), as well as a large increase in DHA (7.5%). It thus appears that vegetable oil ALA can support a small increase in human bloodstream EPA, but cannot support DHA and therefore will not provide a solution to the worldwide shortage of EPA and DHA.

STEARIDONIC ACID AS A SOURCE OF EPA/DHA

Another possible contributor to the EPA/DHA supply for humans is stearidonic acid (SDA, 18 : 4n3). Large commercial concerns have developed a genetically modified soybean that contains a large proportion of this omega-3 fatty acid, and it has been declared safe for human consumption by the US FDA (http://www.fda.gov/ForConsumers/ConsumerUpdates/ucm185688.htm.). The advantage of this approach is that it bypasses the delta-6

Table 3. Alpha-linolenic acid vs. eicosapentaenoic acid vs. docosahexaenoic acid supplementation and human bloodstream content of n-3 fatty acids: Increase in percentage of total erythrocyte n-3 fatty acids after 6 weeks of supplementation

Fatty acid supplemented	ALA diet group	EPA diet group	DHA diet group
ALA	0.24*	−0.05	0.00
EPA	0.12	4.41*	1.56*
DHA	−0.92	−0.90	7.5*
Total omega-3 fatty acids	−0.56	3.38*	9.08*

* Indicates statistically significant difference. ALA, alpha-linolenic acid; DHA, docosahexaenoic acid; EPA, eicosapentaenoic acid.
Data taken from [18].

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desaturase enzyme, often considered a rate-limiting step, and leads to greater conversion to EPA relative to ALA. James and Cleland compared directly the efficacy for increasing long-chain omega-3 fatty acids in the human bloodstream after feeding encapsulated ALA or SDA for 3 weeks, reviewed in [19]. They observed that both plasma and erythrocyte EPA, as well as its elongation product, docosapentaenoic acid (22 : 5n3), were both increased with SDA and more so than with ALA intake; however, there was no increase in DHA. It was quite interesting to note that they observed the relative effectiveness for increasing tissue EPA for supplemental EPA, SDA and ALA was 1 : 0.27 : 0.07, respectively. Harris also performed a similar study and found that SDA was somewhat less efficacious in increasing human erythrocyte EPA than in the study by James as it was only about 17% as effective as was EPA, reviewed in [19]. Again, SDA was unable to increase human bloodstream DHA. Although there is as of yet no current commercial production of this oil, the SDA may in the future have utility in supporting human blood levels of EPA.

ALGAL OILS

What viable alternatives are there then for future increases in EPA and DHA production? One candidate will certainly be heterotrophic microorganisms (Fig. 4) such as algae. Commercial operations have for some time been performed employing *Crypthecodinium cohnii* as a source of DHA for infant formula [20]. Several species of Schizochytrium are also used to make commercial oils for supplements and the food and beverage markets and variants can make mainly DHA or both EPA and DHA [21]. Such production is somewhat capital-intensive, but the raw materials, that is, sugar and various vitamins and minerals, are plentiful and thus production can in principle be potentially increased enormously. At present, algal oil represents less than 2% of human EPA/DHA consumption, but this source has grown appreciably due to several socially desirable properties including its environmental friendliness, the absence of ocean borne contaminants, its vegetarian nature, as well as the possibility to manufacture under kosher or halal conditions. However, fermentation and refining of algal oils is at present more expensive to produce than fish oils and likely to remain so for some time. Thus, microorganism-based production of EPA/DHA-enriched oils can accommodate greater demand for these nutrients albeit at a higher price.

KRILL OIL

Krill oil has been another more recent contributor to the supply of EPA/DHA. The trade association, GOED, estimated krill oil catch (in 2013) was 202 000 tons, which is consistent with an estimate of 212 000 tons given by Kwantas and Grundman [22]. GOED calculated that the total EPA/DHA oil available from this was about 625 tons. Using the figure above of a total human consumption of about 200 000 tons, the krill oil contribution would represent 0.3% of the total. However, the present krill oil harvest is estimated to be below 0.1% of the available biomass, which has been estimated to be more than 300 million tons in the Antarctic Ocean [22]. The Commission for the Conservation of Antarctic Living Resources has estimated that 5.6 million tons could be sustainably harvested [22]. Thus, it appears that the krill oil supply could be increased substantially to a level where it would supply 8–9% of the present human consumption. One would surmise, however, that the whole catch could not come from a particular area, but must be

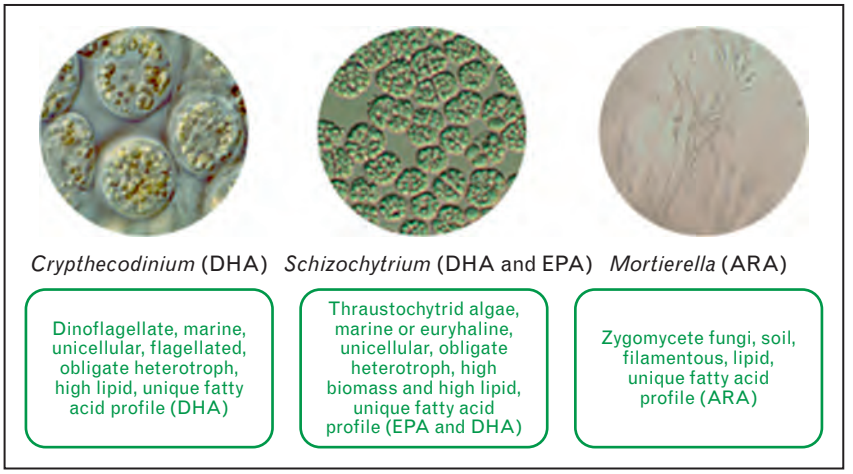


FIGURE 4. Three important commercial organisms used for making long chain polyunsaturated fatty acids for human nutrition.

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spread out over the ecosystem, and this represents a challenge for the industry so as not to disrupt a fragile environment. Krill oil has been marketed as more efficacious than fish oil purportedly due to its differing lipid class form. Some species of krill oils have large proportions of phospholipid forms of EPA/DHA [23], although this is variable. However, the claims of greater bioavailability of krill oil are not founded on controlled clinical studies and have been criticized [24▪].

GENETICALLY MODIFIED VEGETABLE OIL AS A FUTURE SOURCE OF EPA/DHA

Finally, the potential production of EPA and DHA by land-based plants must be considered. Several groups have made considerable progress in producing transgenic oil seed crops that can produce EPA or DHA. Petrie *et al.* [25] have demonstrated insertion of various desaturase and elongase genes into *Arabadopsis thaliana* in order to produce oils containing 12–15% of fatty acids as DHA. They state that one hectare of a *Brassica napus* crop employing this technology producing 12% DHA oilseed would be equivalent to that produced from 10 000 fish. Dow Agro Sciences and DSM Inc. have similarly announced a program to produce DHA in canola oil [26]. Ruiz-Lopez *et al.* [27] have recently demonstrated transgenic variants of *Camelina sativa* that produce seeds with up to 31% EPA and also one with up to 12% EPA and 14% DHA.

Another very interesting development in the genetically modified arena is the introduction of domestic animals that have cloned enzymes such that their meat is enriched in omega-3 fatty acids. An example of this is given by the transgenic pig rich in EPA/DHA after introduction of a humanized *Caenorhabditis elegans* gene, fat-1, which encodes for the omega-3 fatty acid desaturase [28▪]. With this approach, they were able to increase the content of omega-3 fatty acids from 2.2% in wild-type pigs to 8.6% in the transgenic ones, with substantial increases in EPA especially but also in DHA. Production of domesticated animals on a large scale with these genes is a potential significant source of EPA/DHA in the future human diet, although no such commercial enterprise has yet developed. Thus, it seems that the long delayed technical success of transgenic terrestrial plants and animals that contain DHA has finally come. Of course, these products must undergo a lengthy registration with various regulatory bodies in various countries before they can be commercialized. At present, there is considerable resistance to the use of genetically modified foods in the Western world in particular. It may be expected though that with price pressure and a limited supply of these essential nutrients, the genetically modified sources of EPA/DHA will become more widely available in the future.

CONCLUSION

The major sources of EPA and DHA are from fish and fish oils, and there is not nearly enough of it available to supply commonly recommended levels of intake for these nutrients in the human population. Unfortunately, ALA, although relatively abundant, cannot substitute for its longer chain and more unsaturated n-3 family members since humans have only a trace level capacity to perform this conversion *in vivo*. Microbial oil production may theoretically be expanded in an almost unlimited fashion, but at a higher price point. Eventually, lower-priced plant-based DHA oils will become available, but from a genetically modified organism. Economic, social, regulatory and political issues will thus be the main determinants of the availability of EPA and DHA in the future.

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Conflicts of interest

Norman Salem, Jr and Manfred Eggersdorfer are employed by DSM, a manufacturer of omega-3 fatty acids.

REFERENCES AND RECOMMENDED READING

Papers of particular interest, published within the annual period of review, have been highlighted as:

- ▪ of special interest
- ▪▪ of outstanding interest

1. Miller M, Stone NJ, Ballantyne C, *et al.* American Heart Association Clinical Lipidology, Thrombosis, and Prevention Committee of the Council on Nutrition, Physical Activity, and Metabolism; Council on Arteriosclerosis, Thrombosis and Vascular Biology; Council on Cardiovascular Nursing; Council on the Kidney in Cardiovascular Disease. Triglycerides and cardiovascular disease: A scientific statement from the American Heart Association. Circulation 2011; 123:2292–2333.
2. Australia – Position statement. Fish, fish oils, n-3 polyunsaturated fatty acids and cardiovascular health. 2008. www.heartfoundation.org.au.
3. FAO. FAO report of an expert consultation on fats and fatty acids in human nutrition. 2010 Bulletin 91. Report from Joint FAO/WHO Expert Consultation on fats and fatty acids in human nutrition 10–14 November 2008, Geneva, Switzerland.
4. EFSA: Scientific opinion on dietary reference values for fats, including saturated fatty acids, polyunsaturated fatty acids, monounsaturated fatty acids, trans fatty acids, and cholesterol. EFSA J 2010; 8:1461.

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5. Japanese Ministry of Health, Labor and Welfare, 2009. Dietary reference intakes for Japanese (2010 edition). MHLW Website 2010. http://www.mhlw.go.jp/shingi/2009/05/s0529-4.html. [Accessed 14 June 2011]
6. Netherlands – English translation, Health Council of the Netherlands. Guidelines for a healthy diet 2006. The Hauge: Health Council of the Netherlands, 2006; publication no. 2006/21E. http://www.gr.nl/pdf.php?ID=1481andp=.
7. ANZ – reference values for Australia and New Zealand including recommended dietary intakes. 2006. www.nhmrc.gov.au/publications/synopses/_files/n35.pdf.
8. Belgium – Superior Health Council. Recommandations Nutritionnelles Pour La Belgique. CSS No. 8309. Revision 2009.
9. ANSES. l'actualisation des apports nutritionnels conseille?s pour les acides gras. Rapport d'expertise collective. 2011. http://www.anses.fr/Documents/NUT2006sa0359Ra.pdf. [Accessed 16 December 2014]
10. Kuratko C, Salem N Jr. Standards for preventing and treating omega-3 fatty acid deficiency. In: Robert McNamara, editor. The omega-3 fatty acid deficiency syndrome: opportunity for disease prevention. Nova Science Publishers NY; 2013. pp. 399–420.
11. ADA Vegetarian Nutrition Evidence-Based Nutrition Practice Guideline 2011. ADA Evidence Analysis Library. http://www.adaevidencelibrary.com/topic.cfm?cat=4021andauth=1. [Accessed 7 October 2011]
12. FAO. The state of the world fisheries and aquaculture. Rome: Food and Agriculture Organization of the United Nations; 2014.
13. Naylor RL, Hardy RW, Bureau DP, *et al.* Feeding aquaculture in an era of finite resources. PNAS 2009; 106:15103–15110.
14. http://www.ers.usda.gov/data-products/oil-crops-yearbook.aspx. [Accessed December 2013]
15. Goyal A, Sharma V, Upadhyay N, *et al.* Flax and flaxseed oil: an ancient medicine and modern functional food. J Food Sci Technol 2014; 51:1633–1653.
16. Turchini GM, Nichols PD, Barrow C, Sinclair AJ. Jumping on the omega-3 bandwagon: Distinguishing the role of long-chain and short-chain omega-3 fatty acids. Crit Rev in Food Sci Nutr 2012; 52:795–803.
17. Brenna JT, Salem N Jr, Sinclair AJ, Cunnane SC. α–Linolenic acid supplementation and conversion to n-3 long chain polyunsaturated fatty acids in humans. Prost Leuk Essential Fatty Acids 2009; 80:85–91.
18. Egert S, Lindenmeier M, Harnack K, *et al.* Margarines fortified with α–linolenic acid, eicosapentaenoic acid, or docosahexaenoic acid alter the fatty acid composition of erythrocytes but do not affect the antioxidant status of healthy adults. J Nutr 2012; 142:1638–1644.
19. Walker CG, Jebb SA, Calder PC. Stearidonic acid as a supplemental source of w-3 polyunsaturated fatty acids to enhance status for improved human health. Nutrition 2013; 29:363–369.
20. Kuratko C, Abril R, Hoffman J, Salem N Jr. Enrichment of infant formula with omega-3 fatty acids. In: Jacobsen C, Nielsen NS, Horn AF, Sorensen A-DM, editors. Food enrichment with omega-3 fatty acids. Woodhead Publishing Ltd; 2013. pp. 351–386.
21. Kuratko CN, Salem N Jr. Long chain omega-3 fatty acids from algae: composition, stability, metabolism and health effects. Eur J Lipid Sci Technol 2013; 115:965–976.
22. Kwantas JM, Grundmann O. A brief review of krill oil history, research, and the commercial market. J Dietary Supp 2014. doi: 10.3109/19390211.2014.902000.
23. Arujo P, Zhu H, Breivik JF, *et al.* Determination and structural elucidation of triacylglycerols in krill oil by chromatographic techniques. Lipids 2014; 49:163–172.
24. ■ Salem N Jr, Kuratko CN. A reexamination of krill oil bioavailability studies. Lipids Health Dis 2014; 13:137–142.

An important review as it clarifies that krill oil has not been shown to be more bioavailable than fish oil or other sources of EPA/DHA as has been claimed.

25. Petrie JR, Shrestha P, Zhou X-R, *et al.* Metabolic engineering plant seeds with fish oil-like levels of DHA. PLOS One 2012; 7:1–7.
26. Walsh TA, Metz JG. Producing the omega-3 fatty acids DHA and EPA in oilseed crops. Lipid Technol 2013; 25:103–105.
27. Ruiz-Lopez N, Haslam RP, Napier JA, Sayanova O. Successful high-level accumulation of fish oil omega-3 long-chain polyunsaturated fatty acids in a transgenic oilseed crop. Plant J 2014; 77:198–208.
28. ■ Zhou Y, Lin Y, Wu X, *et al.* The high-level accumulation of n-3 polyunsaturated fatty acids in transgenic pigs harboring the n-3 fatty acid desaturase from *Caenorhabditis briggsae*. Transgenic Res 2014; 23:89–97.

A demonstration that it may be possible to introduce genes into domestic animals so that they may convert omega-6 fatty acids to omega-3 fatty acids for subsequent human consumption.

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